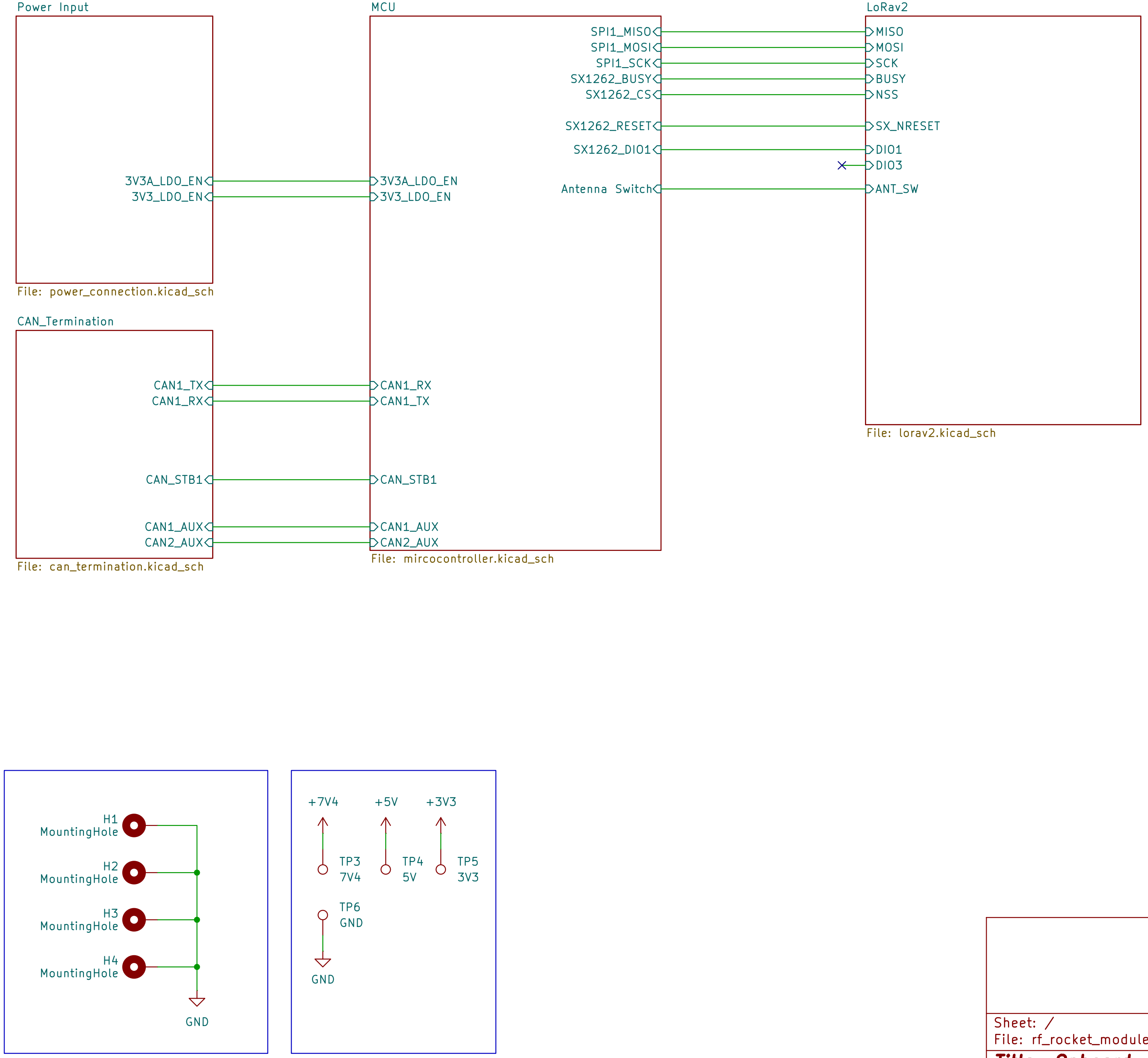

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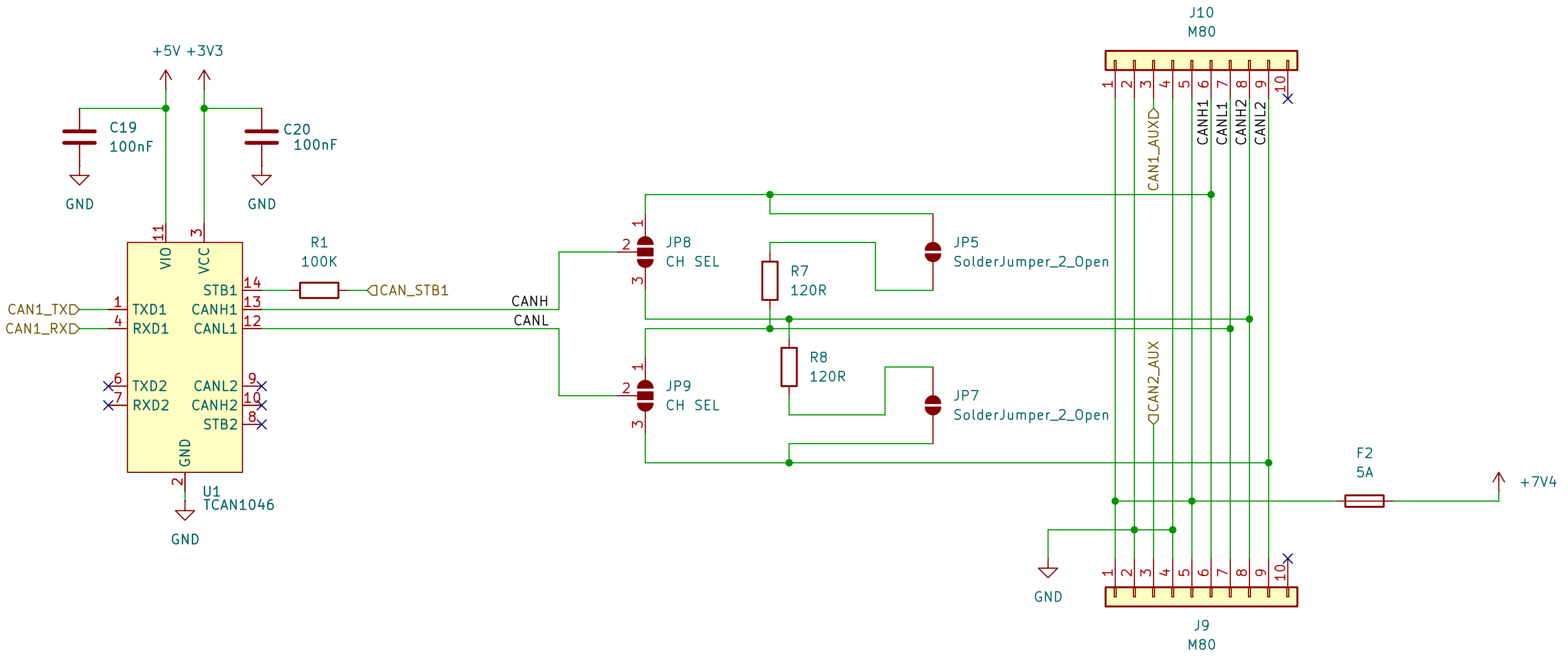
Title: Onboard Radio

Size: A4	Date: 2026-05-28	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 1/5



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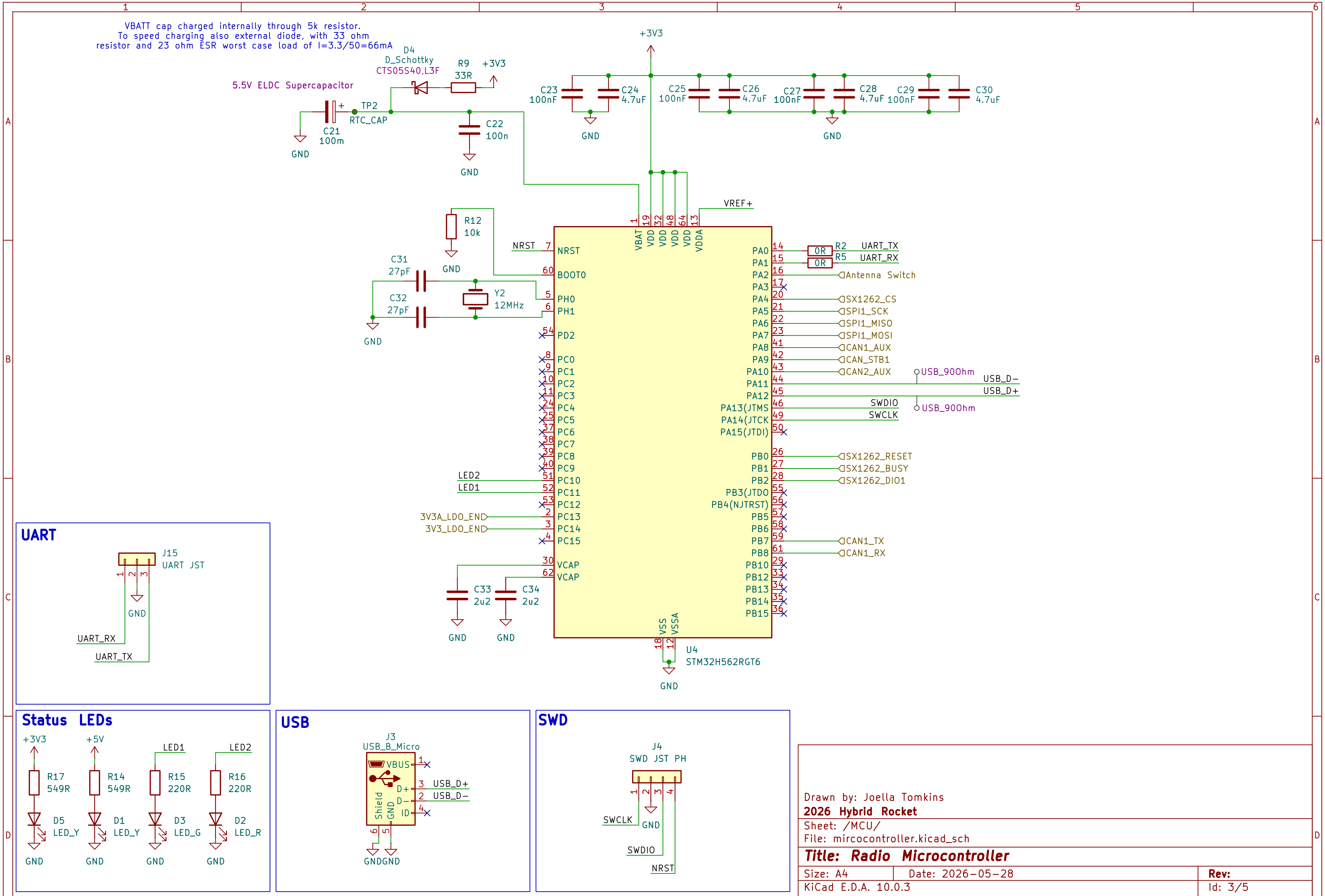
File: can_termination.kicad_sch

Title: CAN Transceiver

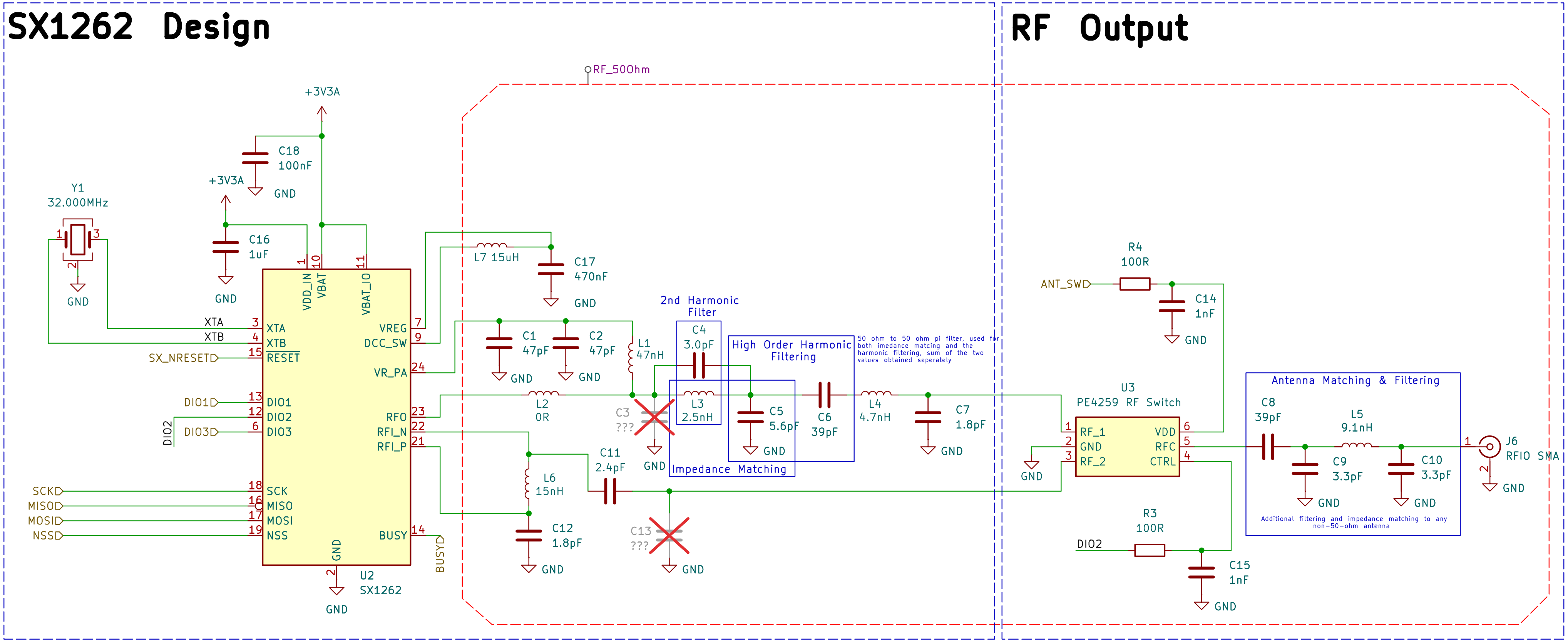
Size: A4	Date: 2026-05-28	Rev: A
KiCad E.D.A. 10.0.3		Id: 2/5



Drawn by: Joella Tomkins
2026 Hybrid Rocket
Sheet: /MCU/
File: mircocontroller.kicad_sch

Title: Radio Microcontroller

Size: A4 | Date: 2026-05-28 | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 3/5



2nd Harmonic Notch – L3 || C4

f = 1 / (2 * pi sqrt(L3 * C4)) = 1838 MHz = 2 * 915MHz

Parallel LC trap tuned to the 2nd harmonic. Present high impedance at 1830MHz to block it, near invisible at 915MHz fundamental.

Primary Match – L3 + C5

L-match solved for Zopt = 13.5+j7.6Ω (Semtech load-pull optimum) -> 50ohm:
Series X = 14.6ohm -> 2.54nH ≈ 2.5nH
Shunt X = −30.4Ω → C = 5.72pF ≈ 5.6pF
Transforms PA's optimum load impedance (13.5+j7.6Ω) up to 50Ω for the antenna switch/output stage.

High-Order Harmonic Filter – C6, L4, C7 (+ C5 shared)

C5 already accounted for by the match above; C6 (39pF) is the added pure-filter capacitance, giving C5+C6 = 44.6pF total shunt.

50Ω pi low-pass section (shared with C5) providing extra attenuation of 3rd+ harmonics beyond the notch filter. Series L4 (27Ω reactance) + shunt C7 (96.6Ω reactance) roll off with frequency.

RX Balun/Match – L6, C11, C12

Half-circuit match of Zopt/2 = 37+j67Ω (differential LNA optimum, per Semtech source-pull) → 50Ω:
Series X ≈ −87Ω → C ≈ 1.96pF ≈ C11/C12 (2.4pF/1.8pF)
Shunt X ≈ 84Ω → L ≈ 14.7nH ≈ L6 (15nH)
Converts single-ended 50Ω line to LNA's differential input impedance (74+j134Ω) while providing balun function (single-ended↔differential).

Switch Bias – R3, R4, C14, C15

XC(1nF) @ 915MHz = 0.17Ω – effectively RF short.

R3/R4 (100Ω) current-limit the switch's CTRL/VDD lines; C14/C15 decouple those DC control lines, shunting any coupled RF to ground.

RFC Choke – L7

XL(15μH) @ 915MHz = 86.2 kΩ

RF choke on VR_PA supply – reactance is high enough at 915MHz to isolate the PA's RF swing from the 3V3A rail while still passing DC bias.

Antenna Matching – C8, L5, C9, C10

Pi network for matching/filtering to the actual antenna impedance (not 50Ω-to-50Ω by default) – values set from measured antenna feedpoint impedance rather than calculated from the SX1262 side.

Unpopulated – L2, C3, C13

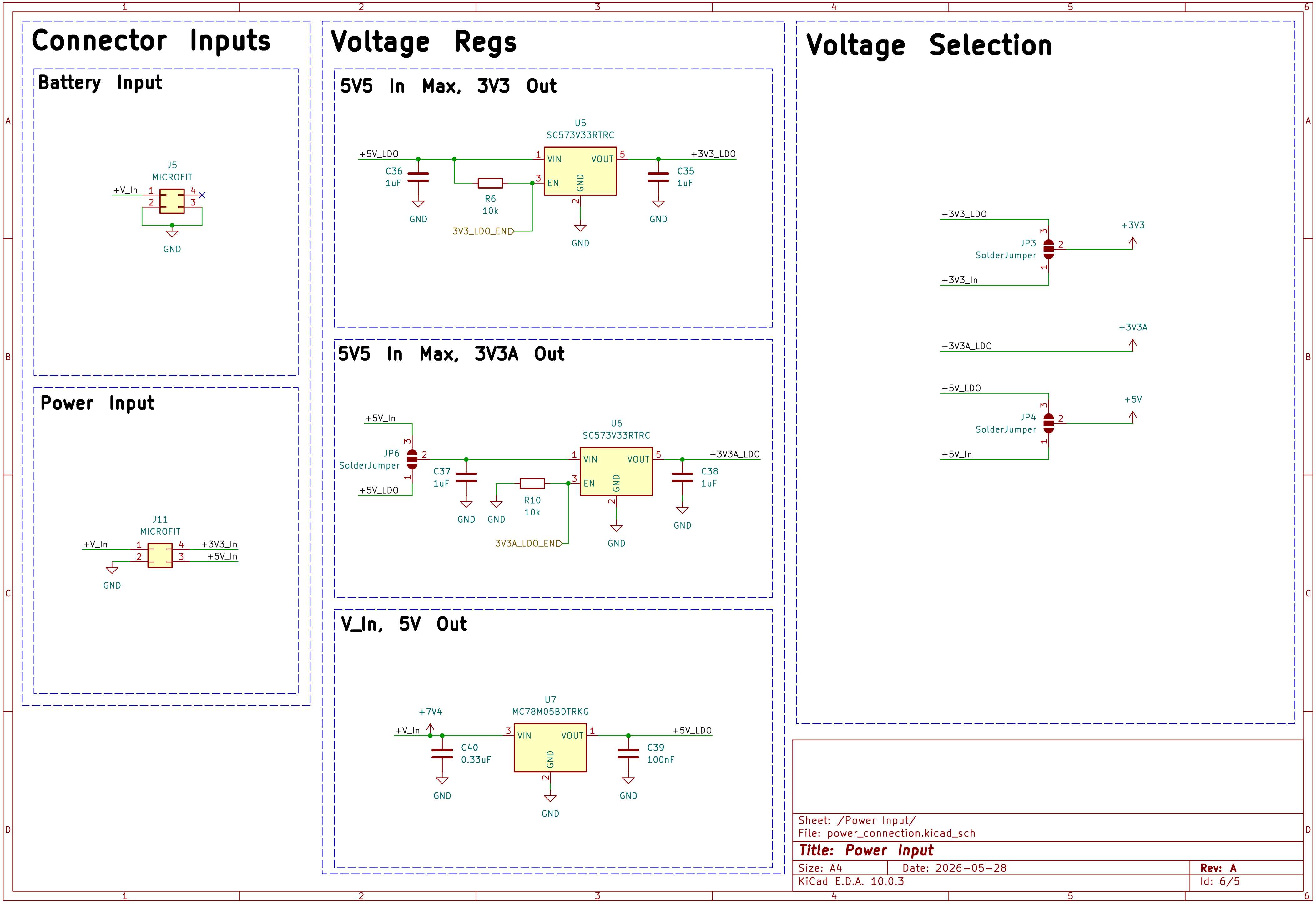
L2 = 0R link (placeholder, no reactance). C3/C13 = spare shunt tuning stubs, DNP – reserved for post-fab VNA tuning if needed; leaving them open doesn't break the signal path.

stm32wl
https://cdn-reichelt.de/documents/datenblatt/A200/SX1262REFERENCE.pdf
https://www.semtech.com/products/wireless-rf/lora-connect/sx1262dvk1cas

Sheet: /LoRav2/
File: lorav2.kicad_sch

Title: Radio IC

Size: A4	Date: 2026-05-28	Rev: A
KiCad E.D.A. 10.0.3		Id: 5/5


Connector Inputs
Battery Input
J5
MICROFIT
+V_In
GND
Power Input
J11
MICROFIT
+V_In
+3V3_In
+5V_In
GND
Voltage Regs
5V5 In Max, 3V3 Out
U5
SC573V33RTRC
VIN
VOUT
EN
GND
+5V_LDO
+3V3_LDO
C36
1uF
R6
10k
C35
1uF
3V3_LDO_END
GND
5V5 In Max, 3V3A Out
U6
SC573V33RTRC
+5V_In
JP6
SolderJumper
+5V_LDO
C37
1uF
R10
10k
C38
1uF
+3V3A_LDO
3V3A_LDO_END
GND
V_In, 5V Out
U7
MC78M05BDTRKG
+7V4
+V_In
C40
0.33uF
C39
100nF
+5V_LDO
GND
Voltage Selection
+3V3_LDO
JP3
SolderJumper
+3V3_In
+3V3
+3V3A
+3V3A_LDO
+5V_LDO
JP4
SolderJumper
+5V_In
+5V
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File: power_connection.kicad_sch
Title: Power Input
Size: A4
Date: 2026-05-28
KiCad E.D.A. 10.0.3
Rev: A
Id: 6/5